Reasoning Failures in LLMS

GROUP 8

SYED ABDULLAH(60104641)

GHULAM SHAH(60106493)

Introduction:

Large Language Models (LLMs) and Vision-Language Models (VLMs) have shown significant progress in solving a wide range of tasks, including natural language understanding, image recognition, and reasoning. However, recent research has highlighted important limitations in their ability to perform reliable reasoning, especially in complex and real-world scenarios.

This project aims to investigate reasoning failures in LLMs and VLMs across three key domains: physics and scientific reasoning, perception-based visual understanding, and visual spatial reasoning. While models often perform well on simple and structured tasks, their performance drops significantly when tasks require multi-step reasoning, precise visual interpretation, or dynamic interaction.

To validate these limitations, we analyzed multiple research papers and conducted our own experiments by testing different models, including ChatGPT, Claude, Gemini, and DeepSeek. We created variations of benchmark-style problems and evaluated model responses to identify common failure patterns

The objective of this report is to systematically examine these failures, compare model performance, and provide insights into why these limitations occur, along with possible **directions for improvement.**

Background:

Physics and reasoning:

Recent research has explored the reasoning limitations of Large Language Models (LLMs) and Vision-Language Models (VLMs) across different domains. In physics and scientific reasoning, studies such as Auto-Bench show that models struggle with multi-step reasoning, interacting variables, and maintaining logical consistency. The MoRA framework addresses these issues by introducing a multi-agent refinement process that identifies and corrects different types of errors.

Table 1: Performance comparison of multiple LLMs. We present the average number of iterations each LLM requires to obtain the correct answer, along with the success rate.

Benchmarking (Setting)	Model used for evaluation (Average Iterations, Successful Rate)				
	Claude-3-5-haiku	Gemini-1.5-pro	Llama-3.1-70b-instruct	Gpt-4o	Qwen2.5-72b-instruct
Social Network (Persons: 3)	(**1**, 20%)	(3, 60%)	(3, 70%)	(2, **100%**)	(2, **100%**)
(Persons: 5)	(∞, 0%)	(5, 50%)	(6, 30%)	**(4, 100%)**	(5, **100%**)
(Persons: 10)	(∞, 0%)	(∞, 0%)	(∞, 0%)	**(8, 30%)**	(∞, 0%)
Chemistry (Nodes: 3; States:3)	(**3**, 25%)	(4, 95%)	(5, 15%)	(4, **100%**)	(4, **100%**)
(Nodes: 3; States: 5)	(6, 5%)	(**4**, 85%)	(**4**, 35%)	**(4, 100%)**	**(4, 100%)**
(Nodes: 10; States: 5)	(∞, 0%)	(11, **15%**)	(∞, 0%)	**(10, 15%)**	(∞, 0%)

Model	Dataset	AO	COT	3-Shot	MORA
Gemma 2 27B	MMLU College	51.11%	73.52%	67.64%	**82.20%**
	MMLU High School	55.93%	77.11%	74.45%	**75.88%**
	PhysicsQA	39.18%	54.59%	59.45%	**70.62%**
	SciEval-Static	60.36%	79.26%	53.04%	**88.76%**
LLaMa 3 70B	MMLU College	59.41%	71.76%	71.76%	**78.82%**
	MMLU High School	60.16%	72.88%	73.66%	**78.81%**
	PhysicsQA	38.37%	56.76%	59.29%	**70.14%**
	SciEval-Static	70.07%	82.23%	63.41%	**86.58%**

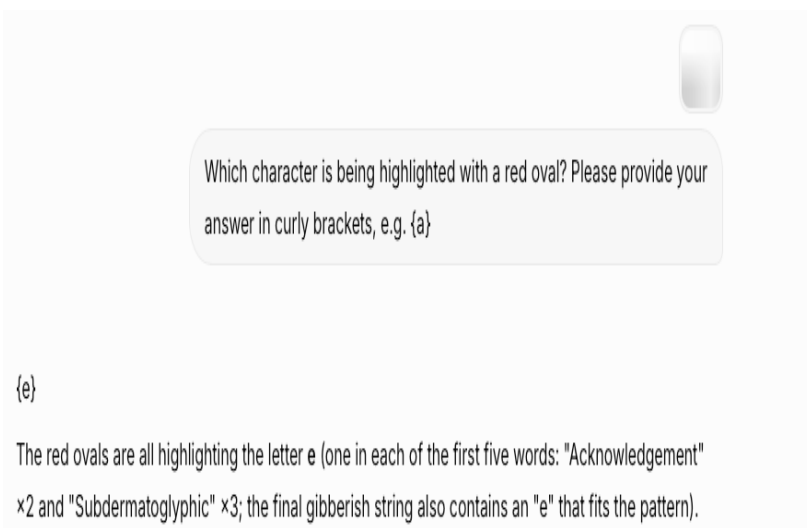
Table 3: Comparison of baseline approaches with MoRA across four datasets: SciEval-Static, PhysicsQA, MMLU High School and College based on final answer accuracy.

Whats the problem with the image?:

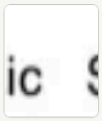
In perception-based tasks, works such as BlindTest and SVAT demonstrate that models fail to accurately interpret fine-grained visual details, including counting objects, detecting overlaps,

and identifying anomalies. Similarly, the ROME benchmark highlights that models can recognize objects but often fail to correctly identify unusual attributes or relationships in counter-intuitive images.

Model	a.	b.	c.	d.	e.	f.	g.	h.	i. Task mean
Random	33.33	50.00	5.77	20.00	20.00	25.00	4.55	33.33	24.00
GPT-4o	41.61	75.91	74.23	41.25	20.21	55.83	39.58	53.19	50.23
Gemini-1.5	66.94	**93.62**	83.29	20.25	24.17	87.08	39.39	53.13	58.48
Sonnet-3	43.41	86.46	72.06	29.79	1.87	65.00	36.17	31.11	45.73
Sonnet-3.5	**75.36**	90.82	**87.88**	**66.46**	**77.71**	**92.08**	**74.26**	**58.19**	**77.84**
Mean	56.84	86.70	79.36	39.44	30.99	74.99	47.35	48.90	58.07



Grok Failed



Which character is being highlighted with a red oval? Please provide your answer in curly brackets, e.g. {a}

Looking at the image, the red ovals are highlighting specific characters in each word:

1. **"Acknowledgement"** (first) — {g}
2. **"Acknowledgement"** (second) — {g}
3. **"Subdermatoglyphic"** (first) — {m}
4. **"Subdermatoglyphic"** (second) — {h}
5. **"Subdermatoglyphic"** (third) — {c}
6. **"tHyUiKaRbNqWeOpXcZvM"** — {b}

If you meant a single specific word, let me know which one!

Claude Sonnet4.6 Failed

Got it — you want **each circled character separately (left → right)** 👀

Looking carefully at the image, the red ovals highlight these characters:

1. In **"Acknowledgement" → e**
2. In **"Acknowledgement" (second) → e**
3. In **"Subdermatoglyphic" → o**
4. In **"Subdermatoglyphic" (next) → p**
5. In **"Subdermatoglyphic" (next) → h**
6. In the random string → **o**

✅ Final Answers (left → right)

👉 **{e}, {e}, {o}, {p}, {h}, {o}**

ChatGpt5 Failed

Only Gemini Passed

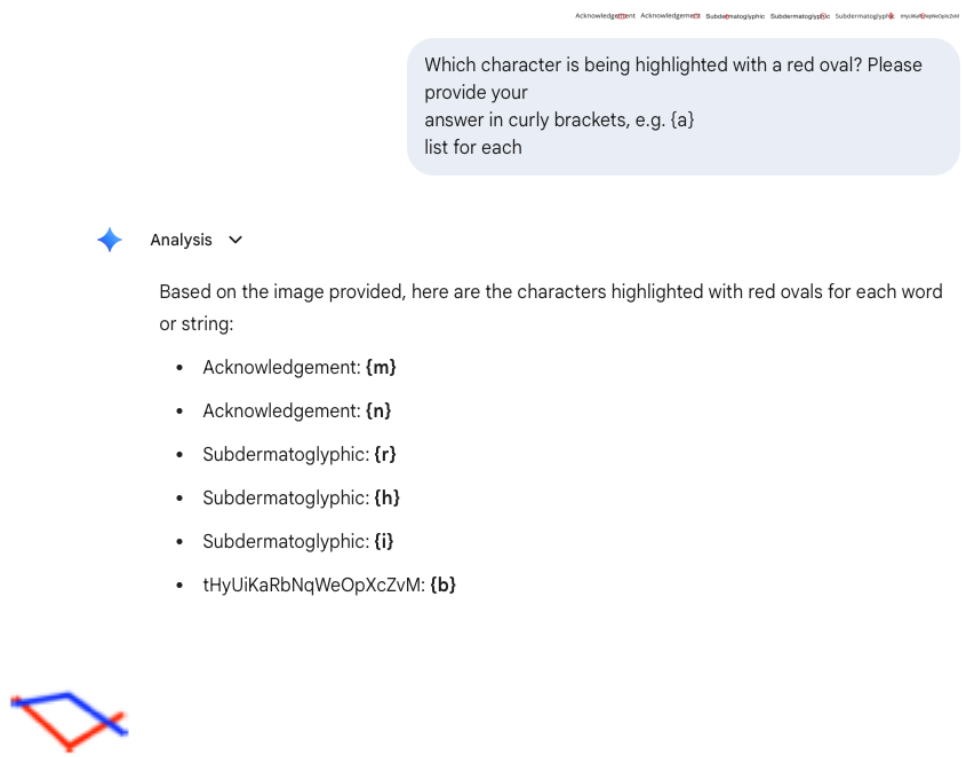
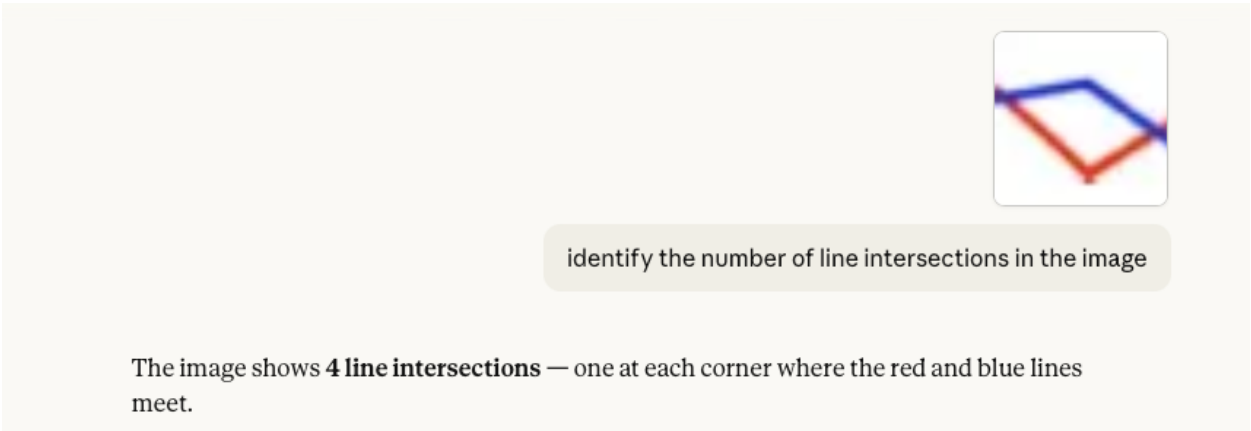


Figure 1



Claude failed

identify the number of line intersections in the image

Looking carefully at the image 👀

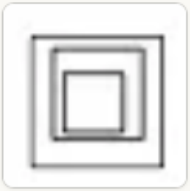
- There are two lines (one blue, one red)
- They cross each other **once**

✅ Final Answer:

👉 **1 intersection**

GPT also failed

Figure 2



count the total number of nested squared

12:56

The image shows **2 nested squares** — one larger outer square and one smaller inner square.

Claude Failed

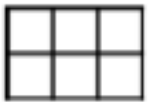
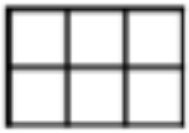


Figure 3



count the number of rows and columns

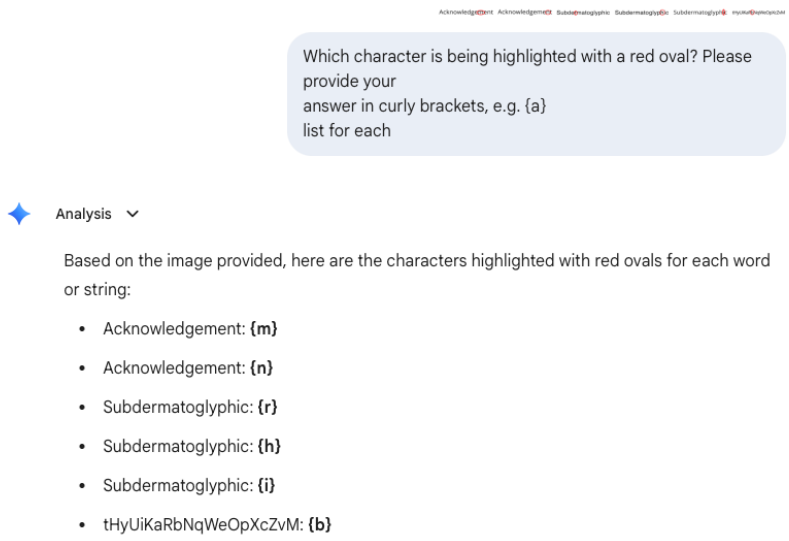
Let's carefully look at the grid 👀

- Horizontally (top to bottom): there are **2 rows**
- Vertically (left to right): there are **4 columns**

✅ Final Answer:

👉 **2 rows and 4 columns**

GPT FAILED



Gemini passed

Figure 4

Grok Failed

Visual Spacial Reasoning:

In visual spatial reasoning, research shows that while models perform well in simple static scenes, they struggle with dynamic tasks involving motion, object interaction, and multi-step planning. These findings suggest that current models lack robust real-world reasoning capabilities, especially when tasks require precise understanding and interaction.

Figure 1: Frames from an example dynamical sequence in our TraySim dataset. The left-most frame shows the first frame of the scene with many objects on the tray and is going to be impacted by a black pusher (right-bottom). The subsequent frames show the state of the system at the 25-th, 50-th, and the 200-th time step (each step is 0.01s). Our task is for the LLM to reason through the dynamics of the system and predict the stability of each object on the tray at the end of the episode, in a zero-shot manner.

In this experiment, the model is evaluated on its ability to perform dynamic spatial reasoning using a sequence of visual frames from a simulated environment. The scene consists of multiple objects placed on a tray, which are then disturbed by an external force (a black pusher). The model is shown several frames representing different time steps of the interaction and is required to reason about how the objects move and interact over time. Based on this temporal sequence, the model must predict the final state of the scene, specifically determining whether each object remains stable or becomes unstable by the end of the episode. This task evaluates the model's ability to understand motion, physical interactions, and long-term outcomes, rather than relying on static image recognition.

which image is to the left of the triangle and how far is it?

Analysis

The image to the left of the triangle is a **red circle**.

In the first image, the shapes are arranged in a horizontal line. The red circle is the first shape on the left, followed immediately by the yellow triangle.

Regarding the distance, it is approximately **one shape-width** away, as the objects appear to be evenly spaced across the center of the frame.

Gemini passed

which image is to the left of the triangle and how far is it?

Great — this is exactly how you should test models 👩‍🏫
Let me answer it clearly like an LLM should:

✅ Answer

👉 The object to the **left of the triangle** is:

Red circle

GPT

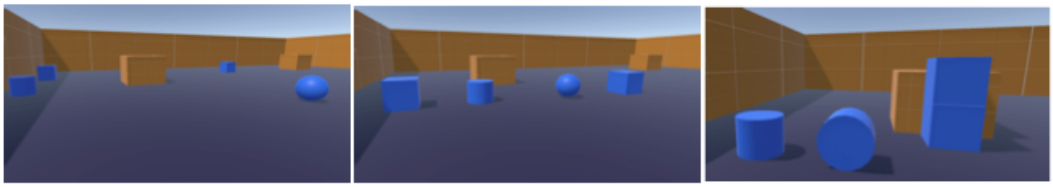


Figure 3: Examples of controls placed on the visual input to LLaVA.

	Free-text		1-sentence	
	Stable	IoU	Stable	IoU
ChatGPT	.33	.50	.67	.75
LLaMA 2-7B	.00	.50	.50	.50
LLaMA 2-70B	.25	.75	1.00†	.33
LLaVA	.50	.75	.50	.75
	No player			
	Free-text		1-sentence	
	Stable	IoU	Stable	IoU
LLaVA	.25	.75	.67	.75

The results presented in Figure 3 highlight the performance of multiple models on embodied spatial reasoning tasks involving object manipulation and physical stability. The evaluation compares different models under two response formats: free-text responses and constrained one-sentence outputs. Performance is measured using stability (correctness of physical reasoning) and Intersection over Union (IoU), which evaluates spatial accuracy.

The findings show that model performance varies significantly depending on both the model architecture and the response format. ChatGPT demonstrates moderate performance, achieving improved stability when constrained to one-sentence responses. LLaMA-based models exhibit inconsistent behavior, with LLaMA 2-70B achieving perfect stability in the one-sentence setting

but lower spatial accuracy. LLaVA, a vision-language model, shows relatively balanced performance across both formats but does not consistently outperform other models.

Additionally, the “no player” setting, where the agent is not actively interacting with the environment, shows reduced performance, indicating the importance of interaction context in spatial reasoning tasks. Overall, the results confirm that current models struggle to reliably perform embodied reasoning tasks, particularly when physical stability and precise spatial understanding are required.